

Running Python

Open a terminal and run:

```
python
```

Basic Operations

```

3 + 2
2 * (3 + 5) * 4
5 / 2
5 // 2 # integer division
5 % 2  # modulo
2 ** 3 # power

```

Variables

```

m = 8
m * 2
a = (1 + 2) * 6

```

Types of Variables

```

greeting = "Hello"
name = "Peter"
greeting + " " + name
# "Hello Peter"

is_man = True
is_woman = not is_man

```

Functions

```

import math

math.cos(0)
math.pow(5, 4)
math.log(10)

```

Define your own function

```

def sum(a, b):
    return a + b

sum(2, 3)

```

Quadratic Equation

Equation: $ax^2 + bx + c = 0$

Solution: $(-b \pm \sqrt{b^2 - 4ac}) / (2a)$

```

root1
import math

def root1(a, b, c):
    t1 = b*b - 4*a*c
    t2 = math.sqrt(t1)
    return (-b + t2) / (2 * a)

root2
def root2(a, b, c):
    t1 = b*b - 4*a*c
    t2 = math.sqrt(t1)
    return (-b - t2) / (2 * a)

```

Execution

```

root1(1, -5, 6)
root2(1, -5, 6)

```

Conditions

```

if condition:
    statements

```

Comparison operators: <, >, <=, >=, ==, !=

```

1 < 5
1 == 5

```

Using variables:

```

if a < 0:
    number = "negativo"

```

Example with Imaginary Roots

```

import math
import cmath

def root1(a, b, c):
    t1 = b*b - 4*a*c
    if t1 >= 0:
        t2 = math.sqrt(t1)
        return (-b + t2) / (2 * a)
    else:
        t2 = cmath.sqrt(t1)
        return (-b + t2) / (2 * a)

```

Example:

```
root1(1, 4, 8) # (-2+2j)
```

Logical Operators

- **and:** and
- **or:** or
- **not:** not

Truth table for AND:

```

0 and 0 = 0
0 and 1 = 0
1 and 0 = 0
1 and 1 = 1

```

Loops

```

for
for i in range(6):
    print(i)

```

```

while
i = 0
while i <= 5:
    print(i)
    i += 1

```

Lists (Arrays)

```

vec = []
vec.append(25)
a = vec[0]

```

First 100 even numbers:

```

vec = []
for i in range(1, 101):
    vec.append(i * 2)

```

Looping Through Lists

```

v = [4, 7, 9]
for e in v:
    print(e * 2)

```

Dictionaries (Objects)

```

my_obj = {}
my_obj["argument"] = 25
a = my_obj["argument"]

```

Define with multiple fields:

```
b = {"argument": 25, "angle": 180}
```

Simple Script Example

Create a file named [program.py](#):

```

def fx(x):
    return x*x - 9

def fdx(x):
    dh = 0.0001
    return (fx(x + dh) - fx(x)) / dh

x = 4
for i in range(8):
    x = x - fx(x) / fdx(x)
    print(x)

```

Run it:

```
python program.py
```